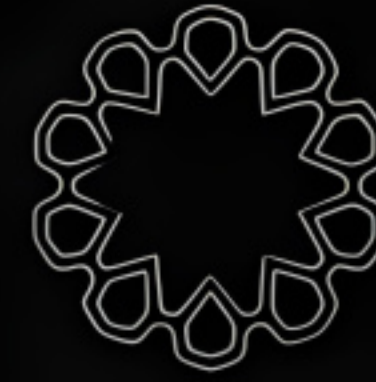




مركز الشيخ مشعل الأحمد الجابر الصباح
للشباب والتكامل السيادي الوطني

HHMYC

DESIGN BASIS REGISTER

REV 1.0
SEPTEMBER 2026



A PLEDGE AND A PROMISE...
FOR THE FUTURE OF KUWAIT

STATUS: GOVERNING REGISTER
CONCEPT STAGE: ALL VALUES SUBJECT TO SURVEY, GEOTECHNICAL

A. GOVERNING GEOMETRY
(PARAMETRIC CONTROL PACKAGE)

ITEM	VALUE
Number of identical units / increment	10 / 36°
Unit outline	Degree-8 Bezier B ₀ -B ₀ , mirrored about U=0.
Unit plan L x W	45.760 x 32.261 m
Floor area / floors	1,050 m² x 3 Levels
FFL	0.000 / +4.800 / +9.200
Roof datum	+13.600

LEVELS

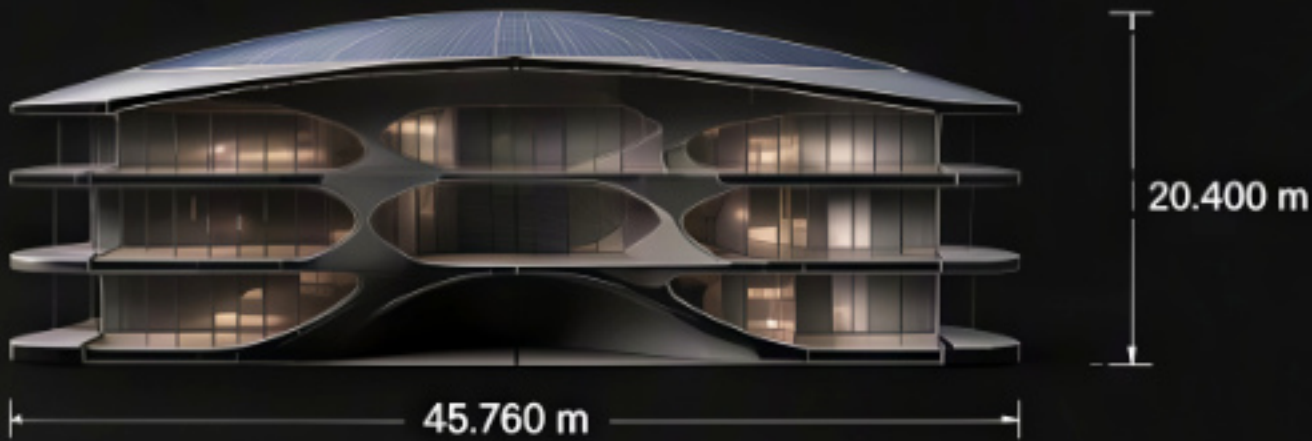
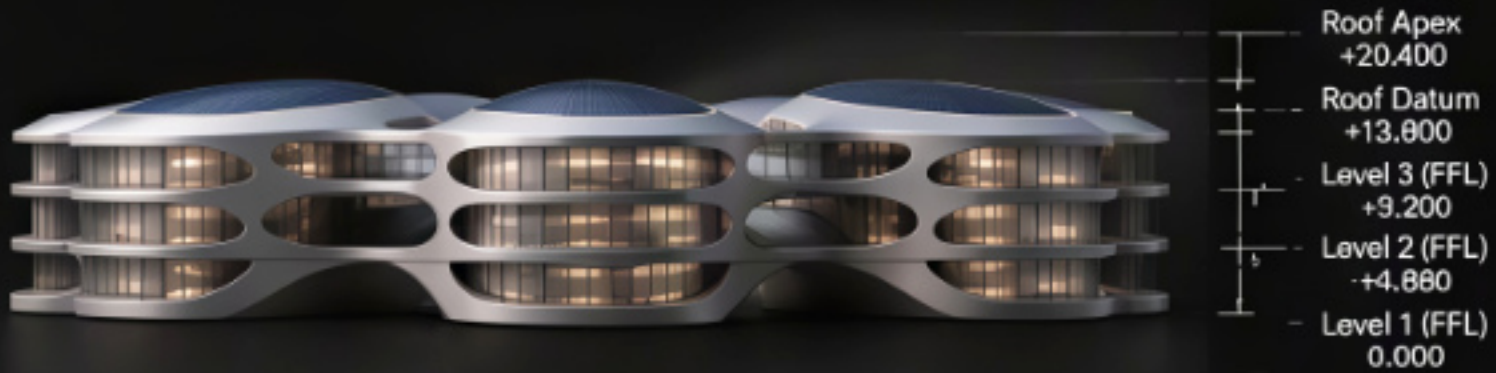
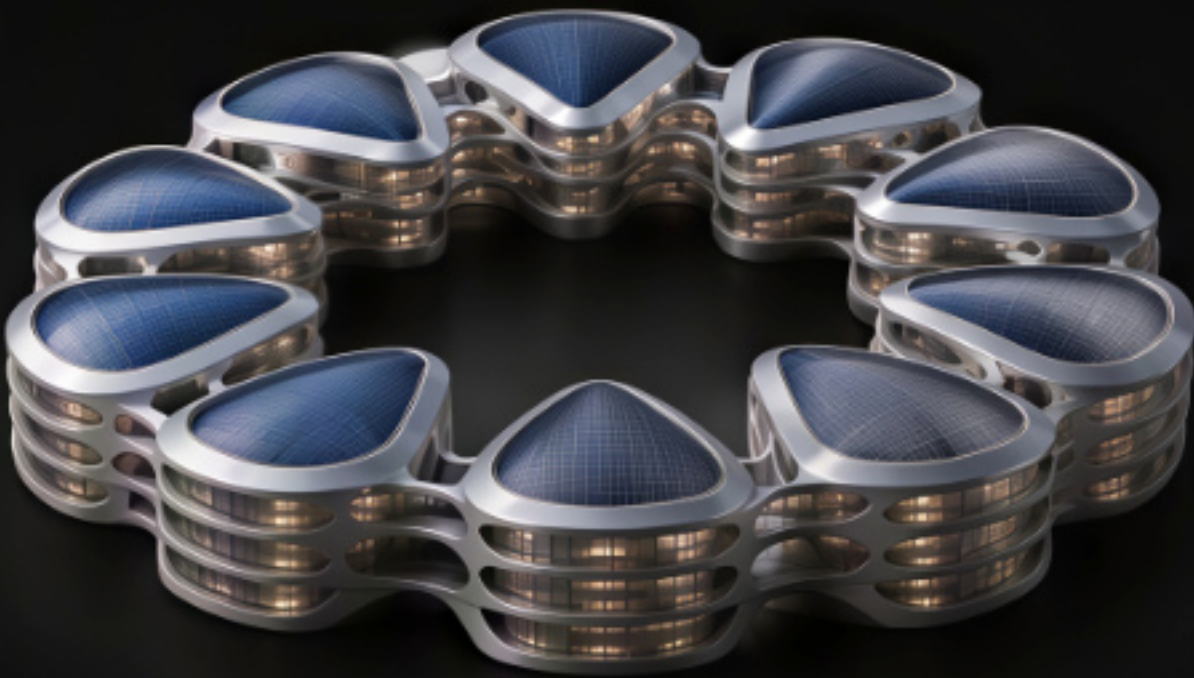
- Roof: Low double-curved gridshell
38.0 x 28.0 m, rise 6.800 m, apex +20.400 m
- Diagrid cell (design basis): 2.4 – 3.0 m
- Minimum plan radii (side/inner/crown):
4.525 / 6.085 / 7.181 m (rule: >= 4.5 m)
- Inner tip radius / oasis diameter:
34.000 m / 68.000 m, fully open
- Outer crown radius / ring diameter:
79.760 m / 159.520 m
- Connectors: 10 per level x 3 levels,
band V=17–23 m, clear gap 6.11–6.74 m,
gross width 6.5 m, 40 m² each.
- Links: 30 necks, 40 m² each.

UNIT DESCRIPTION

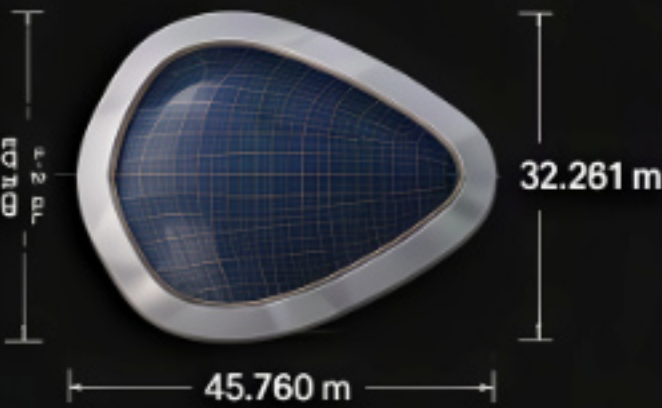
Symmetric soft-edged geometric form.
Not a teardrop, not a petal.

STYLE

Modern, high-end, high-contrast,
black, dark gray.



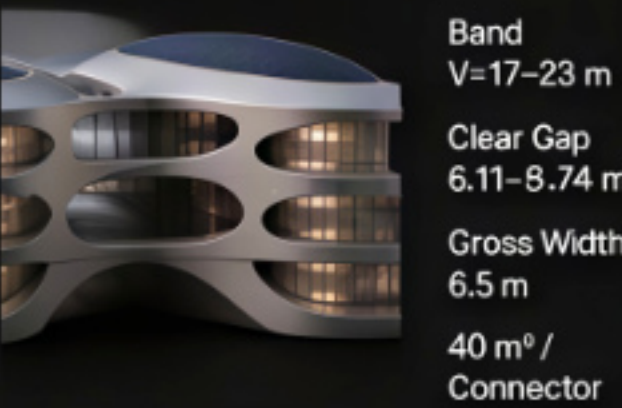
UNIT PLAN



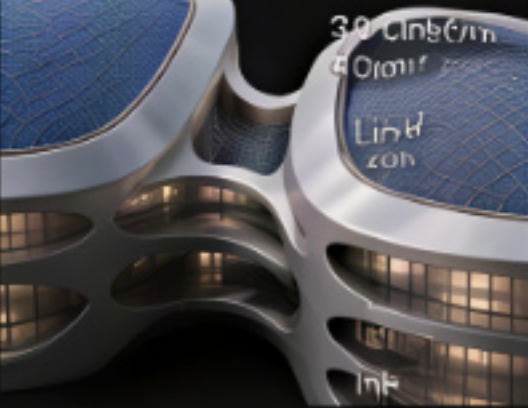
RING GEOMETRY



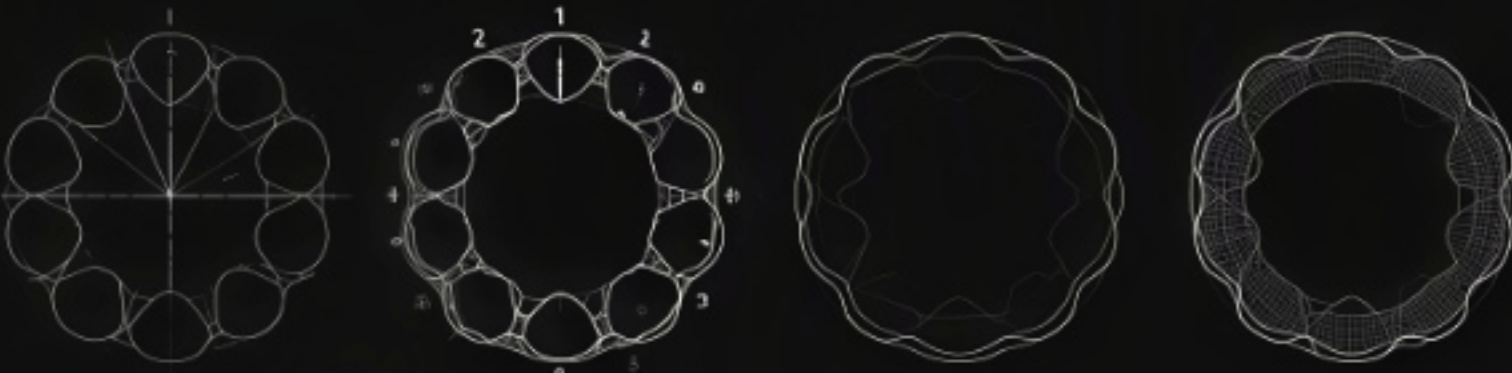
CONNECTOR (LEVEL BAND)



LINK (NECK)



GEOMETRIC LOGIC (10 UNITS / 36°)





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C. AREAS

Total GFA: 46,000 m²

ITEM	VALUE
Ten Units (10 x 3,150 m ²)	31,500 m ²
Connectors (30 x 40 m ²)	1,200 m ²
Hall and Administration Pavilion (800-seat hall, lobbies, admin, hospitality, prayer, clinic, visitor services)	5,500 m ²
Services and Utilities Building (data center, SOC, NOC, BMS, UPS, cooling, substation, generators, batteries, water, security, logistics)	5,200 m ²
Shaded Walkways and Arcades	2,600 m ²
TOTAL SHARED	14,500 m²

KEY PRINCIPLES

- Central oasis: open, unbuilt, not counted in GFA
- No structure or bridge across the oasis
- No basement below sea level
- Service trenches only
- All service and utility buildings located outside the ring
- Raised service platforms above flood datum



SHARED FACILITIES (OUTSIDE THE RING)

- 3 HALL & ADMINISTRATION PAVILION**
5,500 m²
800-seat hall, lobbies, administration, hospitality, prayer, clinic, visitor services

- 4 SERVICES & UTILITIES BUILDING**
5,200 m²
Data center, SOC, NOC, BMS, UPS, cooling, substation, generators, batteries, water, security, logistics

- 5 SHADED WALKWAYS & ARCADES**
2,600 m²
Covered pedestrian networks and arcades connecting shared facilities

OVERALL SITE PLAN



- 1 Ten Units
- 2 Connectors
- 3 Hall & Administration Pavilion
- 4 Services & Utilities Building
- 5 Shaded Walkways & Arcades
- Central Oasis (Open Space)

TYPICAL SECTION (THROUGH UNIT)



RING PLAN DIAGRAM



SITE & INFRASTRUCTURE NOTES

- No basement below sea level
- Service trenches only
- Service trenches only
- All buildings above flood datum
- Resilient infrastructure and redundant utilities



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القبة الرقمية: مسرح الوطن

شاشة بدقة 16K لعرض المحتوى الوطني والتعليمي،
مما يخلق تجربة غامرة دون تكاليف الإضاءة الخارجية.

ARCHITECTURAL DATA

Repetition Angle: 36°
Number of Modules (N): 10
Radial Unit Length: 45.760 m
Unit Max Width: 32.261 m
Unit Start Radius (R0): 34.000 m
Central Opening Diameter: 68.000 m
Outer Crown Radius: 79.760 m
Building Ring Diameter: 159.520 m
Spacing Between Units: 6.11–6.74 m
Architectural Height (no base): 40.8 mm
Total Height (with base): 48.8 mm
Scale: 1:500
Full Print Model Diameter: $\varnothing 560$ mm



DESIGN CONCEPT

A ring of ten modules inspired by the Kuwaiti emblem, united in symmetry to symbolize youth, unity, and vision. Each teardrop opens toward the inner courtyard, creating a seamless flow between culture, learning, and innovation.



SUSTAINABLE
PERFORMANCE



DIGITAL EXPERIENCE



NATIONAL IDENTITY



Primary Structure

Coated carbon steel (A992 / Q355 equivalent), composite deck, braced cores; radial axes R0–R7 at 6 m.



Stainless / Duplex

Exposed elements only (rails, external stairs, spray-zone details).



Roof Gridshell

CHS members with cast/laser-cut nodes on a ring beam at +13.6; ladder-fabricated, ground-assembled, single lift.



BIPV Glazing

Laminated 8+PV+8 mm (max ~20 mm).



Shipping

All kits in 40-foot containers; connections disassembled; marine delivery.



Repetition

One master kit $\times 10$; prototype unit 01 complete before serial.



Steel Tonnage (Benchmark)

Not fixed. Benchmark: 2,600–3,700 t total; manufacturer's ROM governs.

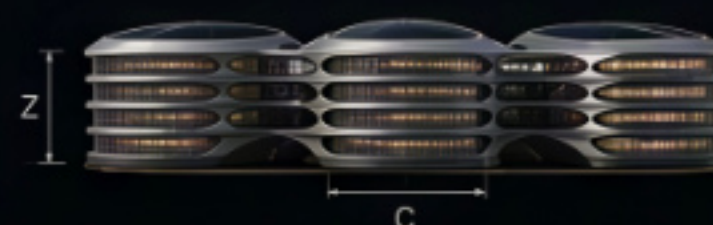


Foundations / Piles

To be determined by geotechnical investigation; no pile length or count is adopted.

GEOMETRIC LOGIC

- Repeating ring module
- $N = 10$
- $R = \text{Variable}$
- $Z = \text{Variable (height)}$
- $T = \text{Variable (thickness)}$
- $C = \text{Variable (opening)}$



MODULE SHAPE (TEARDROP)



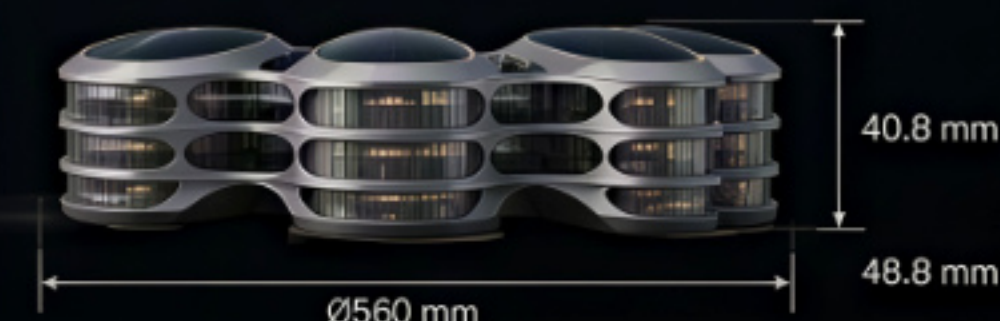
FORM GENERATION ALGORITHM

- 1 Create a ring of radius R
- 2 Divide into N equal segments
- 3 Place teardrop modules tangentially
- 4 Connect with smooth bridges
- 5 Apply thickness T and set height Z
- 6 Ensure interstitial opening C



SCALE & MODEL INFORMATION

Scale: 1:500
Full Print Model Diameter (with base): $\varnothing 560$ mm
Architectural Height (no base): 40.8 mm
Total Height (with base): 48.8 mm

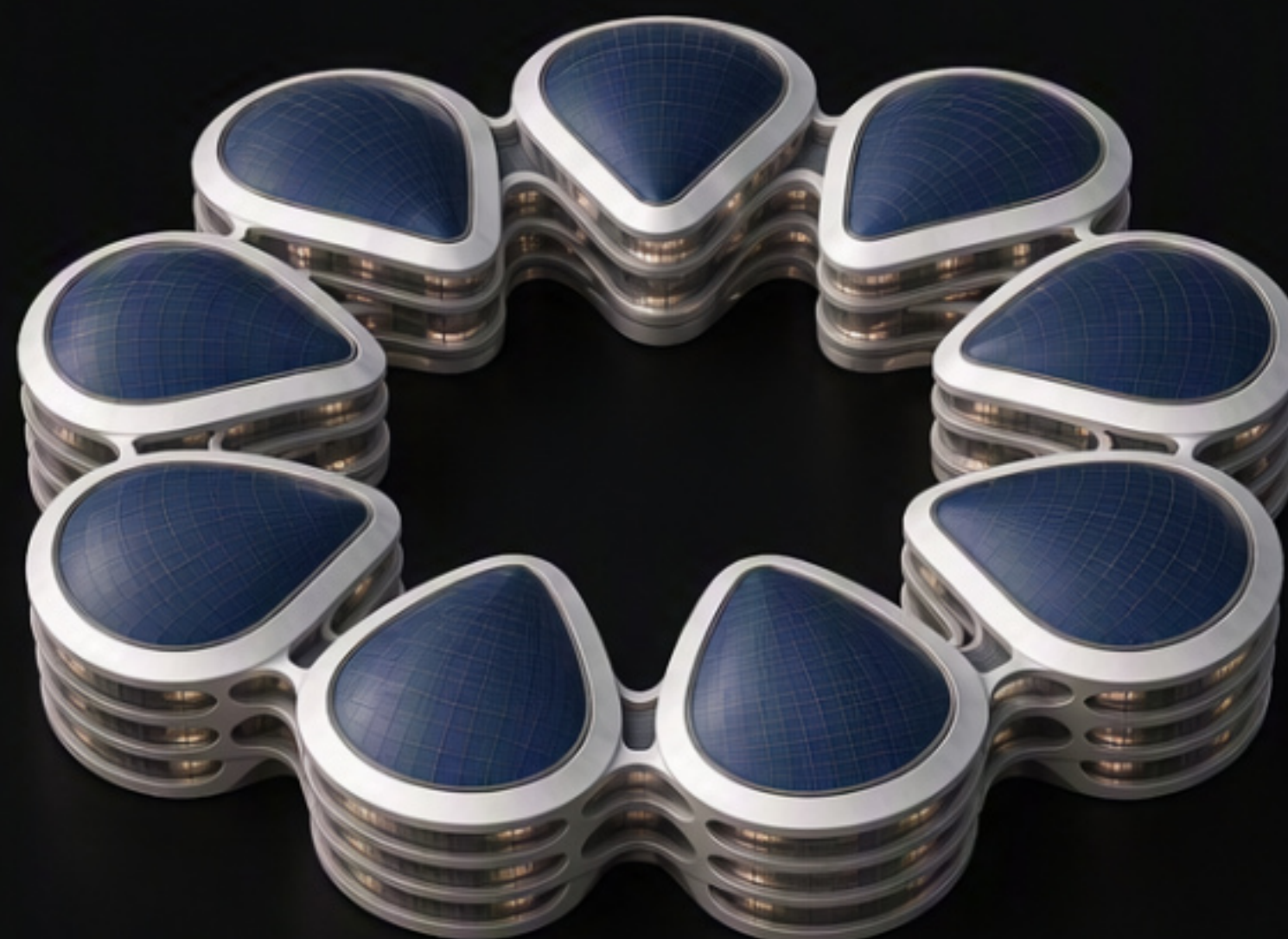




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SHEIKH MISHAL AL-AHMAD AL-JABER AL-SABAH
YOUTH & NATIONAL SOVEREIGN INTEGRATION CENTER

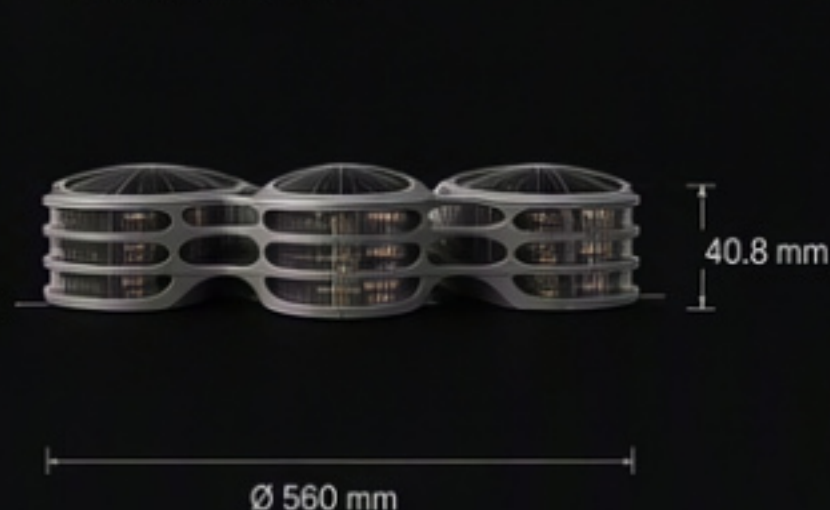
ARCHITECTURAL | SUSTAINABLE | SOVEREIGN



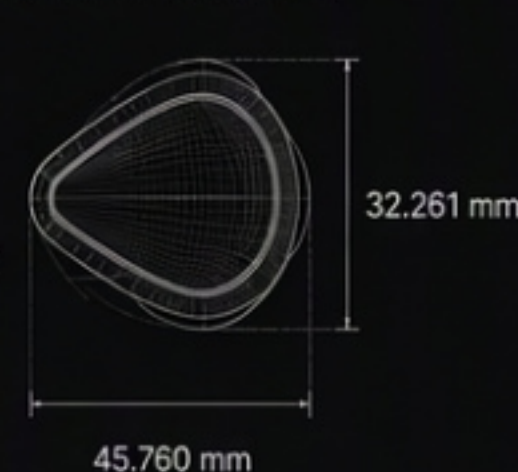
TOP VIEW



FRONT ELEVATION



MODULE (TEARDROP)



E. ENERGY, COOLING, DATA



BIPV ROOFS

1.2 – 1.5 MWp
(about 9,000 m² at 130–160 W/m²)
Optional facade PV up to +0.5 MWp



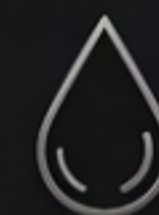
COVERAGE CLAIM

20 – 35% of consumption (to be modelled)
Never state 60–80%



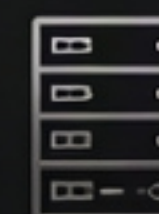
SEAWATER COOLING

Intake from open sea beyond
the breakwater,
titanium exchangers;
lagoon is not the intake



WATER

MEW network supply
+ grey-water reuse;
no desalination line



DATA CENTER

Design target Tier IV or III per OPR;
never "certified" before certification;
above flood datum, in the
services building



VENDORS IN SOVEREIGN / SECURITY SCOPE

Specification-based,
vendor-neutral;
Suppliers named only in



RENEWABLE ENERGY

BIPV integrated roofs
with high-efficiency
photovoltaic glass.



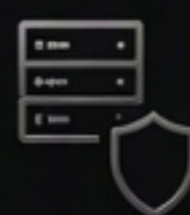
SEAWATER COOLING

Direct intake from
open sea beyond the
breakwater.



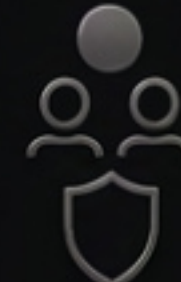
RESOURCE EFFICIENCY

Grey-water reuse
and efficient water
management.



SOVEREIGN DATA

Secure, resilient data
infrastructure above
flood datum.



NATIONAL INTEGRATION

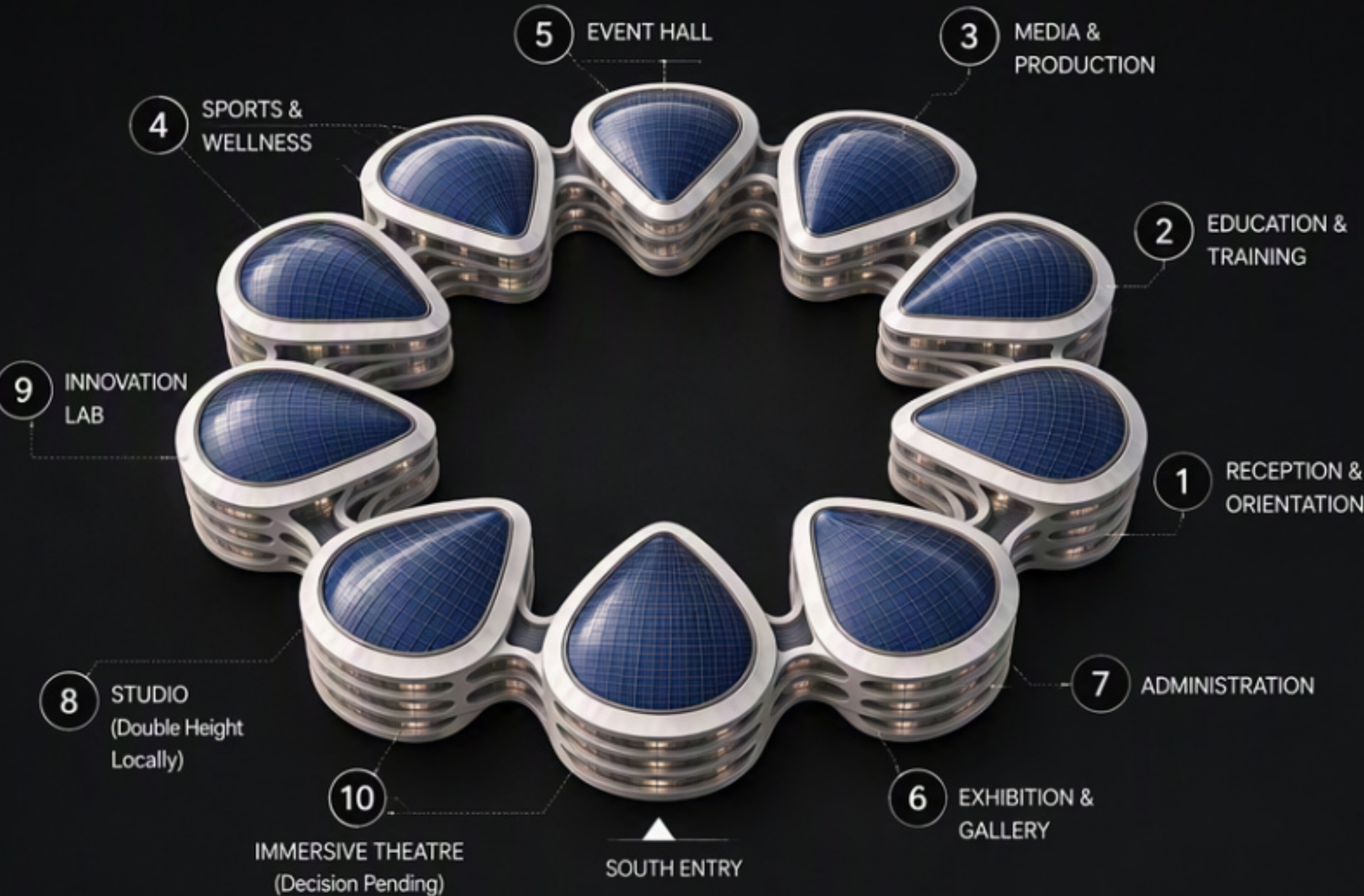
A platform for youth
empowerment and
sovereign capability.

F. PROGRAM

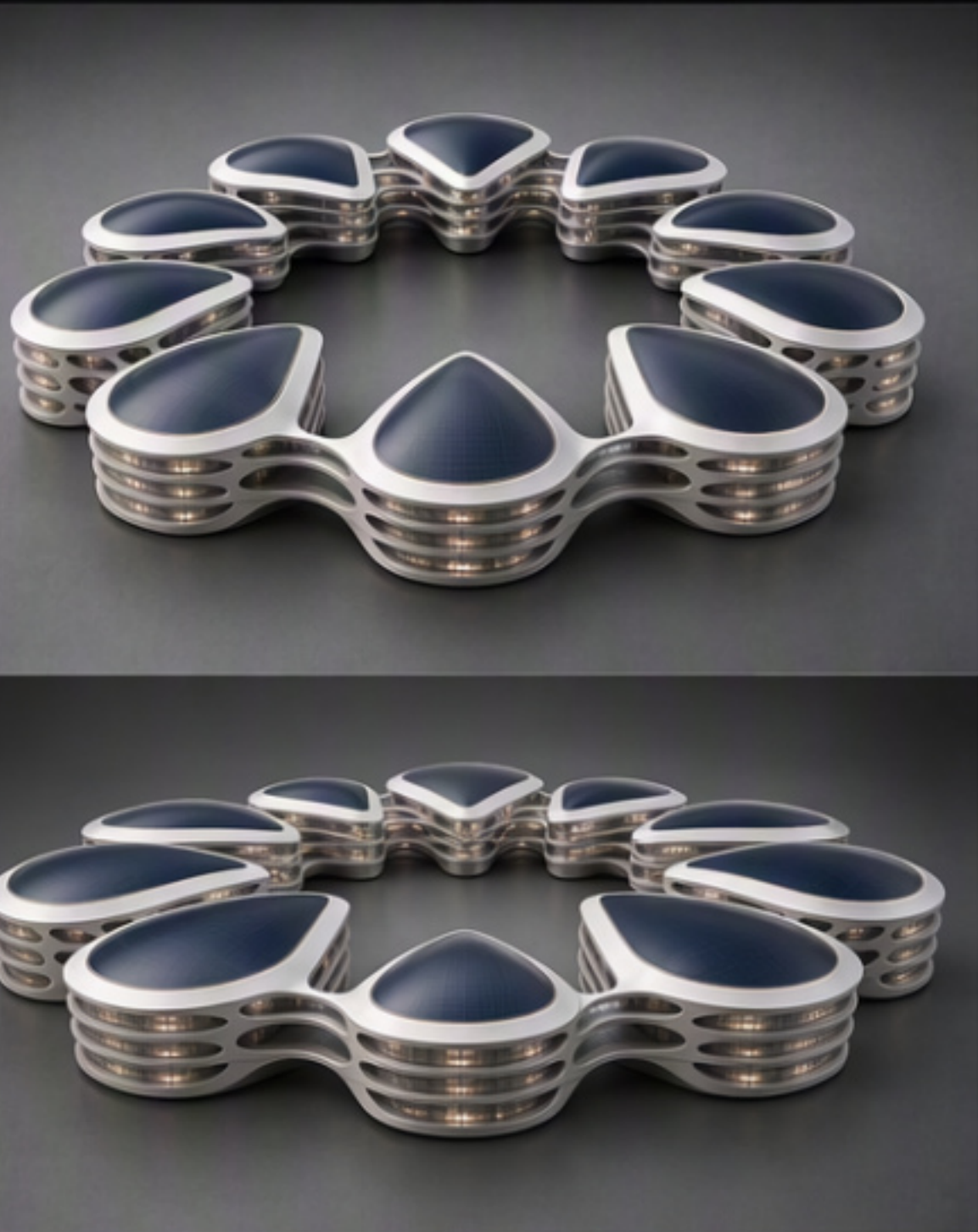
FUNCTIONAL PROGRAM

PROGRAM DATA	
ITEMS (UNITS)	10
CLOCKWISE FROM SOUTH ENTRY	10, 8, 9, 4, 5, 3, 2, 1, 7, 6
CONCURRENT CAPACITY	3,470 in units
SITE PEAK CAPACITY	5,000 – 5,500 (subject to evacuation modelling)
IMMERSIVE THEATRE	Decision pending: small dome 12–14 m in a double-height volume of unit 10, or distributed digital experience. No 20–25 m dome.
STUDIO HEIGHT (UNIT 8)	Local double height by omitting part of level 1 slab; levels remain identical for all units.

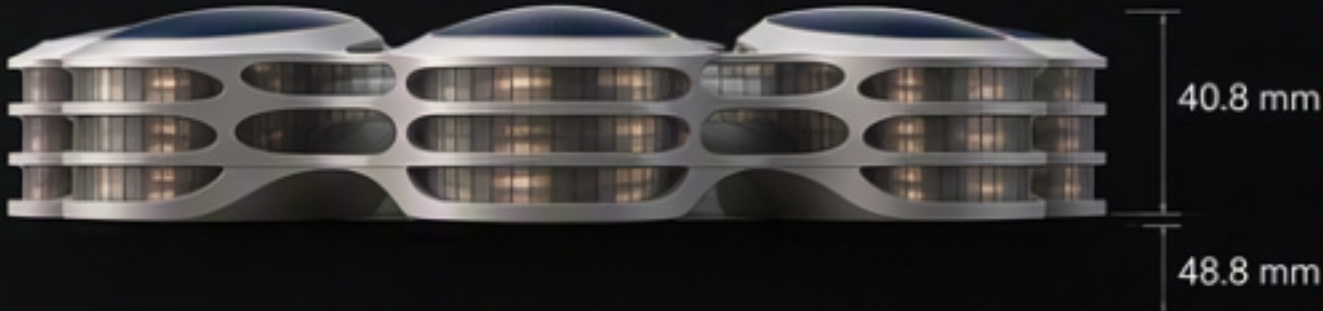
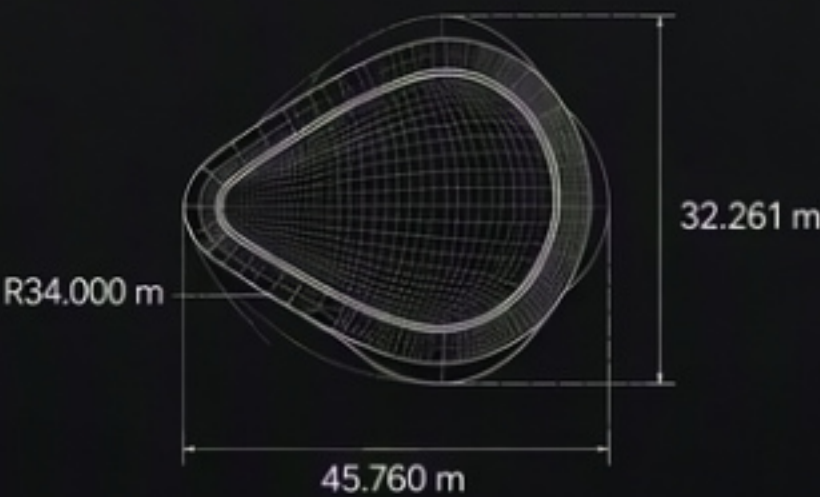
PROGRAM LAYOUT – CLOCKWISE FROM SOUTH ENTRY



ARCHITECTURAL MODEL VIEWS



TECHNICAL DATA	
SCALE	1:500
FULL PRINT MODEL DIAMETER (WITH BASE)	Ø560 mm
ARCHITECTURAL HEIGHT (WITHOUT BASE)	40.8 mm
TOTAL HEIGHT (WITH BASE)	48.8 mm



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TECHNICAL SPECIFICATIONS & COMPONENTS

REV 1.1 | SEPTEMBER 2026

CONCEPT DESIGN, NOT FOR CONSTRUCTION



عهدٌ ووعدٌ ... من أجل الكويت الغد

A PLEDGE AND A PROMISE...

FOR THE KUWAIT OF TOMORROW



التكامل السيادي
الوطني

NATIONAL SOVEREIGN
INTEGRATION



تمكين الشباب
وتطوير القدرات

YOUTH EMPOWERMENT
& CAPACITY BUILDING



الابتكار والتقنيات
المتقدمة

INNOVATION &
ADVANCED TECHNOLOGIES



الاستدامة والمسؤولية
المجتمعية

SUSTAINABILITY &
SOCIAL RESPONSIBILITY



مركز الشيخ مشعل الأحمد الجابر الصباح
للشباب والتكامل السيادي الوطني

ZONE LEGEND

REV 1.1

Each color marks a construction/systems zone in project drawings.
Materials and grades below are based on Rev 1.1.

كل لون يمثل منطقة إنشاء/أنظمة
في الررصمات المشروع.
لا لمواد والدرجات أذناه تعتمد علي Rev 1.1..

ZONE المنطقة	DESCRIPTION الوصف	TYPICAL ELEMENTS العناصر النموذجية	PRIMARY MATERIALS المواد الأساسية	MATERIAL GRADE / SPEC الدرجة / المواصفة	
01	STRUCTURAL FRAME الهيكل الإنشائي	- Main structural system - Primary load-bearing elements	- Columns - Beams - Core walls - Transfer structures	- Reinforced concrete - Structural steel	- Concrete: C40/50 - Rebar: Grade 500 (B500B) - Steel: S355
02	BUILDING ENVELOPE غلاف المبنى	Exterior envelope and weather protection	- Façade panels - Glazing - Roof systems - Waterproofing	- Aluminum system - Low-E glass - Membrane waterproofing	- Aluminum: 6063-T5 - Glass: Low-E, IGU - Membrane: APP/SBS
03	MEP SYSTEMS أنظمة الميكانيكا والكهرباء والسياسة	Mechanical, electrical and plumbing systems	- HVAC - Electrical - Plumbing & Drainage - Fire Protection	- Copper // PEX / uPVC - Galvanized steel - Fire-rated materials	- Copper: ASTM B88 - Steel: ASTM A653 - Pipes: BS EN / ASTM
04	INTERIOR FINISHES التشطيبات الداخلية	Interior surfaces and finishes	- Walls - Floors - Ceilings - Joinery	- Gypsum board - Porcelain / Natural stone - Timber // HPL	- Gypsum: EN 520 - Porcelain: ISO 13006 - Stone: EN 1469
05	LANDSCAPE & EXTERNAL المشيد الخارجي والموقع العام	External works and site development	- Paving - Planting - Irrigation - Site furniture	- Natural stone / Concrete - Soil & planting mix - HDPE // Drip systems	- Stone: EN 1341 - Concrete: C25/30 - Irrigation: ISO 9261
06	SPECIAL SYSTEMS الانظمة الخاصة	Advanced and specialized systems	- BMS - Security - Audio/Visual - ICT / Data	- Fire-rated cabling - Control systems - AV equipment	- Cables: IEC 60332 - BMS: BACnet / KNX - ICT: ISO/IEC 11801

NOTE: CONTRACTOR TO VERIFY ALL MATERIAL GRADES AND SPECIFICATIONS PRIOR TO PROCUREMENT AND CONSTRUCTION.

ملاحظة: يجب على المفاول التحقق من جميع درجات المواد والمواصفات قبل الشراء والتنفيذ.



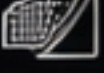
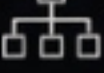
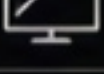
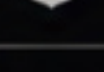
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للشباب والتكامل السيادي الوطني

VENDOR AND PACKAGE RESPONSIBILITY MATRIX

NAMED SUPPLIERS APPEAR ONLY IN NON-SENSITIVE SCOPE.

IN SOVEREIGN, SECURITY, NETWORK, AND DATA-CENTER SCOPES,
SELECTION IS SPECIFICATION-BASED AND VENDOR-NEUTRAL,
DECIDED AT THE FORMAL RFQ STAGE.

REV 1.0 | SEPTEMBER 2026

SCOPE	SCOPE NAMED / BENCHMARK	STATUS	PRIMARY PACKAGE / SCOPE OWNER	VENDOR DESPONSIBILITY/ (RACI)	PACKAGE DESCRIPTION / NOTES
 Primary structural steel	Baosteel (BSN, PRC)	OK	Main Works Contractor	R (Design), A (Quality), C (PMC), I (Client)	High-strength steel, mill certs, traceability.
 Roof gridshell + Diagrid	Zhejiang Southeast Space Frame (PRC)	OK	Main Works Contractor	R, A, C, I	Parametric fabrication, hot-dip galvanizing.
 Third-source diversification / quality benchmark	Hyundai Steel (KOR)	OK	Main Works Contractor	C (Benchmark), A (Quality), I	Benchmark & alternate supply assurance.
 Facade GFRP panels	Facade panels Benchmark: Rieder, Fibrobeton	INFO	Facade Subcontractor	R (Supply), A (Quality), C (PMC), I	GFRP panels with anchorage system.
 Insulating and laminated glass	Insulating and laminated glass Benchmark: Guardian, AGC, Saint-Gobain	INFO	Facade Subcontractor	R, A (Quality), C, I	IGU with low-E, laminated safety glass.
 BIPV roof panels	BIPV roof panels Benchmark: Onyx Solar, LONGi BIPV, ML System	INFO	MEP / Renewable Contractor	R, A (Performance), C, I	Integrated PV with waterproofing system.
 District cooling equipment	Central cooling equipment Benchmark: titanium exchangers per E-Section	INFO	MEP Contractor	R, A (Performance), C, I	Chillers, titanium heat exchangers, pumps.
 Data center Tier IV/III (target)	Data center VENDOR-NEUTRAL specification	OPEN	Data Center Contractor	R, A (Compliance), C (Security), I	Tier III/IV design, Uptime Institute aligned.
 Networking (LAN/WAN/backbone)	Network VENDOR-NEUTRAL. Huawei/Nokia sovereign scope: EXCLUDED	REJ	ICT / Network Contractor	R, A (Becurity), C (Soverpignty), I	Sovereign-compliant equipment & software.
 SOC / cybersecurity	Cybersecurity VENDOR-NEUTRAL. Kaspersky: EXCLUDED (US-sanctioned)	REJ	Cybersecurity Contractor	R, A (Security), C (Compliance), I	24/7 SOC, SIEM, threat intelligence.
 Fire protection	Fire protection Benchmark: Honeywell, Tyco, Siemens	INFO	Fire Protection Contractor	R, A (Compliance), C, I	Sprinklers, gas suppression, FM-Approved.
 Elevators	Elevators Benchmark: KONE, Mitsubishi, Schindler	INFO	Lift Contractor	R, A (Safety), C, I	High-speed lifts, destination control.
 AV, display walls, interactive	Media and screens Benchmark: Samsung, LG, Barco, disguise	INFO	AV / Media Contractor	R, A (Integration), C, I	LED walls, projection, interactive systems.
 Water treatment	Water Treatment Benchmark: Veolia, SUEZ, ACWA	INFO	Water Treatment Contractor	R, A (Performance), C, I	RO, UF, UV, greywater reuse systems.
 Marine works (reclaimed island)	Marine works Benchmark: Boskalis, Van Oord, Jan De Nul	INFO	Marine Contractor	R, A (Marine Compliance), C, I	Reclamation, breakwater, dredging.

LEGEND: **OK** = Confirmed **INFO** = To be Tendered **OPEN** = Specification Released / RFQ Pending **REJ** = Excluded / Not Compliant R = Responsible | A = Accountable | C = Consulted | I = Informed



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TECHNICAL SPECIFICATIONS & PERFORMANCE INDICATORS

REV 1.1 | SEPTEMBER 2026

CONCEPT DESIGN, NOT FOR CONSTRUCTION

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A PLEDGE AND A PROMISE...

FOR THE KUWAIT OF TOMORROW

PERFORMANCE INDICATORS

METRIC	VALUE	BASIS
BIPV roof area (10 units)	~9,000 m ²	Ten low double-curved shells, 38 × 28 m each
Projected BIPV peak capacity	1.2 – 1.5 MWp	At 130–160 W/m ² panel efficiency
Optional facade PV	up to +0.5 MWp	Vertical BIPV on selected facades
Coverage of consumption	20 – 35%	To be modeled post energy simulation
Steel tonnage benchmark	2,600 – 3,700 t	Benchmark only. Manufacturer ROM governs
Roof shell rise-to-span Roof height/span ratio	0.18	6.8 m rise on 38 m span. NOT a pointed dome
Building ring diameter Building ring diameter	159.520 m	10 identical units at 36 degrees
Central open oasis Open oasis Ø	68 m	No structure, no bridges, no service arc
Data center design target Design target	Tier IV or III	Design target, NOT certified before certification
Cooling intake Cooling intake	Open sea, below thermocline	NOT from the lagoon; titanium plate exchangers
Water source Water source	MEW network	No desalination. Grey-water reuse. Zero-Discharge phrasing prohibited
Concurrent capacity in units Concurrent capacity	~ 3,470 persons	Per functional program
Site peak capacity Site peak capacity	~ 5,000 – 5,500	Subject to evacuation modelling
CapEx Capital cost range per phase, not a single figure.	Per Design Basis Register	Open item H-6



التكامل السيادي
الوطني

NATIONAL SOVEREIGN
INTEGRATION



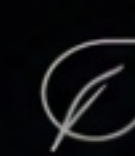
تمكين الشباب
وتطوير القدرات

YOUTH EMPOWERMENT
& CAPACITY BUILDING



الابتكار والتقنيات
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INNOVATION &
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الاستدامة والمسؤولية
المجتمعية

SUSTAINABILITY &
SOCIAL RESPONSIBILITY

All values are concept-design indicators only and subject to change.
Ranges reflect current assumptions and available information.
Detailed engineering, simulation, and certification will refine final figures.

4. APPLICABLE STANDARDS

A. KUWAIT (MANDATORY)

STANDARD / REFERENCE		APPLICABILITY COMPLIANCE
	Kuwait Building Code (KBC) – as amended	Mandatory for all construction and architectural works
	Kuwait Fire Force (KFF) Life Safety Code	Mandatory – Fire, Smoke and Evacuation
	MEW / R-6 (2014) Energy Conservation Code	Mandatory – Energy Efficiency
	Kuwait Environment Public Authority (KEPA) – EIA	Mandatory – Environmental Assessment before Construction
	Kuwait Municipality coastal-strip regulations	Mandatory – Coastal Strip Height Restriction; Special Classification for Sovereign Projects
	Public Tenders Law – Law 49/2016	Mandatory upon government tender

B. INTERNATIONAL (DESIGN REFERENCES)

STANDARD / REFERENCE		APPLICABILITY COMPLIANCE
	ISO 9001:2015 / 14001:2015 / 45001:2018	Quality, Environment and Safety
	ISO 3834-2:2021	Welding Quality
	AISC 360 (steel design, reference)	Steel design – reference
	AWS D1.1 (structural welding code)	Welding of steel structures
	Eurocode 3 (EN 1993) as alternative design ref	Steel design – alternative reference
	GB 50017 (Chinese steel structure code) – fabrication	Fabrication reference
	ASHRAE 90.1 / 62.1 – HVAC and IAQ	HVAC, Energy Efficiency and Indoor Air Quality
	NFPA 13, 72 92 – sprinklers, alarm, smoke control	Fire – as design reference unless otherwise specified
	Uptime Institute Tier III/V – data center (target)	Data center resilience and availability – target
	LEED v4 BD+C – target certification	Sustainability – target certification



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C. EXPLICITLY NOT APPLICABLE

STANDARD / REFERENCE		APPLICABILITY COMPLIANCE
	Estidama Pearl	Not applicable to this project



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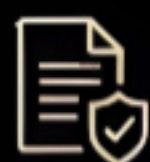
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